

SIMATS ENGINEERING

ASSESSMENT TOOL 1: Agent Design Assessment

COURSE NAME: ARTIFICIAL INTELLIGENCE AND EXPERT SYSTEMS COURSE CODE: MLA01

COURSE OUTCOME COVERED

CO1: Apply search algorithms and heuristic techniques to solve state-space search and optimization problems. (BTL 3)

Assessment Tool Weightage: 55 %

Total Marks: 50

Time: 60 Mins

No. of Questions: 1

Annexure A Agent Design Assessment

Code of Conduc

I Maniyam Anushka(192525411)(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:Maniyam Anushka

Question	Problem Understanding & AI Concept Identification	Dialogflow Agent Design & Implementation	Problem Formulation & Conversation Flow Design	Application of AI Techniques & Reasoning	Testing, Evaluation & Documentation	Total
Weightage	20%	30%	20%	20%	10%	100%
Q1						

Signature of the Faculty

Total Marks Awarded

Scenario

A university wants to develop an **AI-based Student Support Assistant** to provide instant responses to student queries related to course registration, timetable, examination schedules, attendance details, and campus services.

As an AI developer, design and implement a conversational agent using **Dialogflow** that can understand user requests, process intents, and provide appropriate responses.

Q. No.	Criteria to be evaluated	BL	Marks
1	Problem Analysis and Agent Design: Analyze the student support problem and explain how Artificial Intelligence can solve the problem. Design the conversational agent by defining its objective, users, environment, inputs, outputs, and expected performance measures.	BL3	10
2	Dialogflow Agent Development: Create a Dialogflow agent by designing appropriate Intents, Entities, Training Phrases, Responses, and Contexts. Explain how the agent perceives user queries and generates responses. Provide screenshots of the implementation.	BL5	10
3	Problem Formulation and Decision Flow: Represent the chatbot interaction as a problem-solving process by defining the initial state, user goal, possible actions, state transitions, and goal completion conditions. Draw the conversation flow diagram.	BL6	10
4	Search Strategy Application: Analyze how search techniques used in AI problem solving can support conversational agents. Apply a suitable search approach (such as Breadth First Search or heuristic-based search) to model intent selection or dialogue navigation and justify your choice.	BL5	10
5	Testing and Evaluation: Test the Dialogflow agent with different user queries. Evaluate accuracy, response quality, handling of unknown queries, and suggest improvements.	BL6	10

Evaluation Rubrics

Criteria	Weightage (%)	Excellent (Full Marks)	Good	Average	Poor
Problem Understanding & AI Concept Identification	20%	10 – Clearly analyzes the student support problem, defines AI application, agent objective, users, environment, inputs, outputs, and performance measures with proper justification.	7.5 – Identifies most components of the agent design with minor omissions.	5 – Provides basic problem analysis with limited explanation of agent components.	0–2.5 – Incomplete understanding of problem and agent design.
Dialogflow Agent Design & Implementation	30%	10 – Successfully creates a functional Dialogflow agent with appropriate intents, entities, training phrases, responses, contexts, and explains interaction flow with screenshots.	7.5 – Develops major Dialogflow components with minor configuration errors.	5 – Creates basic intents and responses with limited entity/context utilization.	0–2.5 – Incomplete or incorrect Dialogflow implementation .
Problem Formulation & Conversation Flow Design	20%	10 – Clearly represents chatbot interaction using initial state, goal state, actions, state transitions, goal conditions, and a well-designed conversation flow diagram.	7.5 – Provides a structured flow with minor missing details.	5 – Provides basic flow representation with limited explanation.	0–2.5 – Poor or missing problem formulation and decision flow.
Application of AI Techniques & Reasoning	20%	10 – Correctly analyzes and applies suitable search strategies (BFS/heuristic search) for intent selection or dialogue navigation with strong justification.	7.5 – Applies appropriate search strategy with reasonable explanation.	5 – Provides basic understanding of search strategy application.	0–2.5 – Unable to relate search concepts with conversational agent.
Testing, Evaluation & Documentation	10%	10 – Performs comprehensive testing using multiple user queries, evaluates accuracy, response quality, unknown query handling, and suggests meaningful improvements.	7.5 – Performs adequate testing and provides evaluation with minor gaps.	5 – Conducts limited testing with basic observations.	0–2.5 – Insufficient testing and evaluation.

1. Problem Analysis and Agent Design

Problem Analysis

Every semester, university students raise large numbers of repetitive questions about course registration, class timetables, examination schedules, attendance percentage, and campus services (hostel, library, transport). Handling these manually through the administrative office causes long waiting times, especially during registration week and the exam period, and support staff repeat the same answers many times a day.

Artificial Intelligence can solve this by using Natural Language Understanding (NLU) to read a student's free-text question, classify it into a known category of request (an **intent**), pull out useful details such as a course name or date (**entities**), and instantly generate the correct answer — 24 hours a day, without waiting for a human staff member to be free.

Agent Design (PEAS Framework)

The conversational agent can be described using the standard PEAS (Performance, Environment, Actuators, Sensors) framework used in AI agent design:

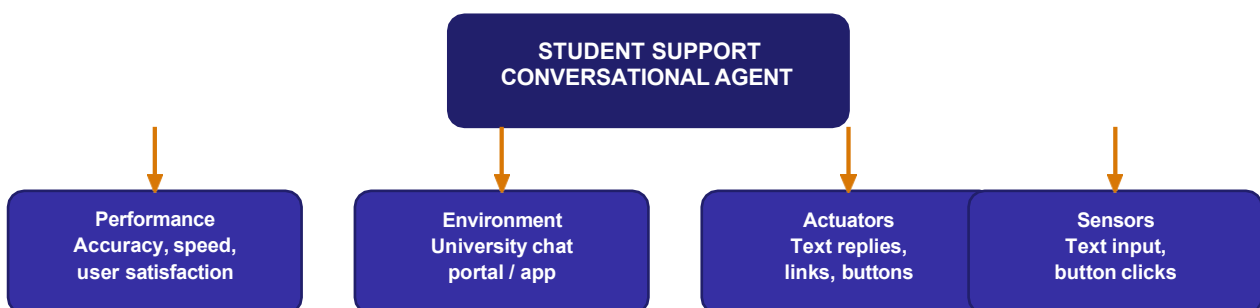


Figure 1: PEAS description of the Student Support Assistant agent.

- **Objective:** Instantly and correctly answer student queries about registration, timetable, exams, attendance and campus services.
- **Users:** Enrolled university students (and optionally staff, for administrative queries).
- **Environment:** A university chat portal or mobile app; the environment is *partially observable* (the agent only sees the typed text, not the student's full situation), *dynamic* (timetables and exam dates change every semester), and *multi-agent* (many students interact with it at once).
- **Inputs:** Typed natural-language text or quick-reply button taps from the student.
- **Outputs:** Text responses, links, and buttons offered back to the student.
- **Performance measures:** Intent-matching accuracy, response time, percentage of queries resolved without human help, and student satisfaction rating.

Based on this design, the agent behaves as a **goal-based agent with internal state**: it keeps track of the ongoing conversation (using Dialogflow *contexts*) so that it can ask follow-up questions, such as "which semester?", before giving a final, goal-satisfying answer.

2. Dialogflow Agent Development

The agent perceives a student's message as raw text, matches it against training phrases for each intent using Dialogflow's built-in NLU model, extracts any entities present, and returns the response configured for that intent (optionally after asking for missing details). The layout below is an illustrative mock-up of

how the intents, entities, training phrases, responses, and contexts are configured inside the Dialogflow console for this agent (a real project would include actual console screenshots at this point).

Dialogflow ES — Student Support Assistant (illustrative layout)

INTENTS

Default Welcome

CourseRegistration.Query

Timetable.Query

ExamSchedule.Query

Attendance.Query

CampusServices.Query

Default Fallback

Goodbye

TRAINING PHRASES

"When is my exam for DBMS?"

"Show my class timetable"

"How do I register for a course?"

"What is my attendance in AI?"

ENTITIES

- @course-name (custom)
- @semester (custom)
- @sys.date (system)
- @sys.person (system)
- @service-type (custom)

RESPONSE

"Your DBMS exam is on 12 Aug 2026, 10:00 AM, Hall B. Anything else?"

CONTEXTS

- Input: awaiting-semester
- Output: exam-followup
- Lifespan: 2 turns

Figure 2: Illustrative layout of the Dialogflow console for the Student Support Assistant (mock-up — replace with real screenshots when the agent is built in the Dialogflow console).

Intents and Sample Training Phrases

Intent Name	Sample Training Phrases	Sample Response
Default Welcome Intent	"Hi", "Hello", "I need help"	Greets the student and lists available help topics.
CourseRegistration.Query	"How do I register for a course?", "Add DBMS to my semester"	Explains the registration steps and portal link for the given semester.
Timetable.Query	"Show my class timetable", "What classes do I have on Monday?"	Returns the timetable for the requested day/semester.
ExamSchedule.Query	"When is my DBMS exam?", "Show exam dates for semester 5"	Returns date, time and hall for the requested exam.
Attendance.Query	"What is my attendance in AI?", "Am I eligible to write the exam?"	Returns attendance percentage and eligibility status.
CampusServices.Query	"Library timings", "Hostel contact number"	Returns the requested campus service information.
Default Fallback Intent	(any unmatched input)	Politely asks the student to rephrase or offers to connect to a human agent.

Entities and Contexts

- **System entities used:** @sys.date (exam/timetable dates), @sys.person (student name).
- **Custom entities:** @course-name (e.g. DBMS, AI, DSA), @semester (e.g. Semester 5), @service-type (e.g. library, hostel, transport).
- **Input/Output contexts:** used to remember information across turns — for example, after asking for the exam schedule, an *exam-followup* context is set with a lifespan of 2 turns, so a follow-up question like "and for DSA?" is understood without repeating the semester.

3. Problem Formulation and Conversation Flow Design

The chatbot interaction can be represented as a classic AI problem-solving / state-space process:

- **Initial state:** Student opens the chat; no intent identified yet.
- **Goal state:** Student's query has been correctly answered (or handed off to a human if the agent cannot help).
- **Possible actions (operators):** Detect intent, ask a clarifying question (slot filling), fetch the requested data, generate a response, end the conversation, or restart the intent-detection step for a new query.
- **State transition:** Each user message moves the conversation from one state (e.g. "awaiting semester") to the next (e.g. "ready to answer") based on the recognised intent and entities.
- **Goal completion condition:** The requested information has been delivered and the student either asks a new question (loop back) or ends the chat.

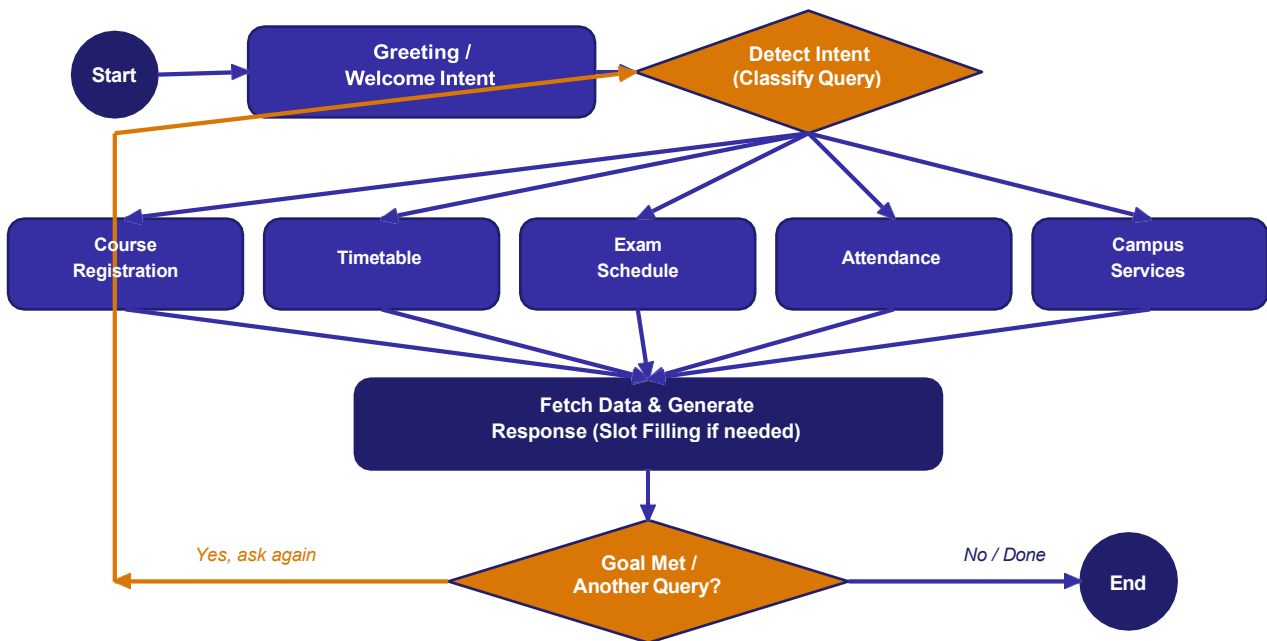


Figure 3: Conversation flow diagram for the Student Support Assistant, from greeting to goal completion.

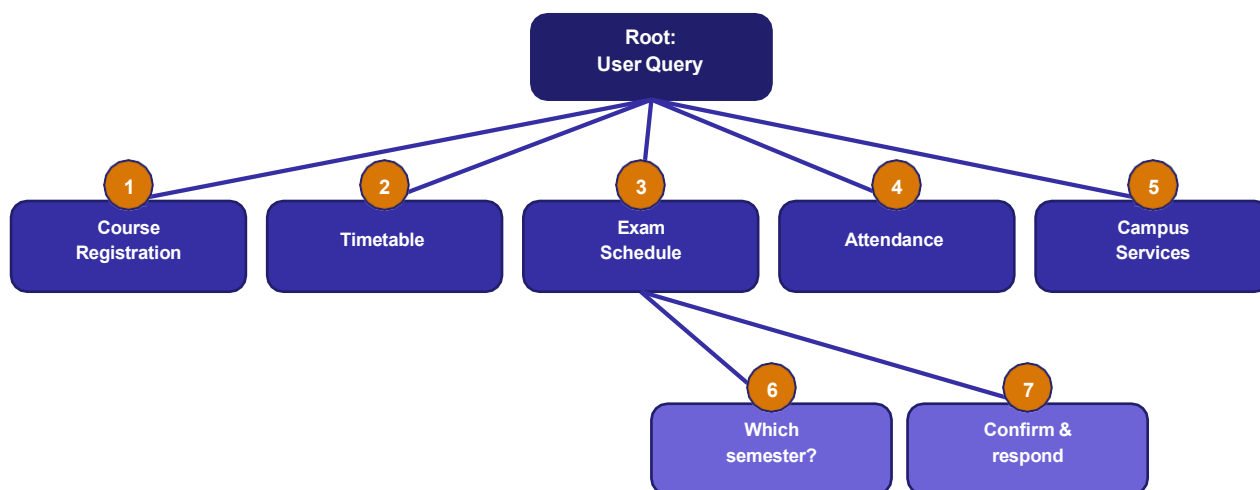
4. Search Strategy Application

Selecting the correct intent for a student's message, and deciding which follow-up question to ask next, can both be modelled as a search problem over a tree of possible conversation states: the root is the newly received query, and each branch represents a candidate intent or a possible next dialogue state.

Why Breadth First Search (BFS) fits intent selection

- The intent tree is **shallow and unweighted** — a query typically maps to one of a small, fixed set of top-level intents, then at most one or two follow-up slot-filling steps. BFS naturally explores level by level, which matches this shallow structure well.
- BFS is **complete** and guarantees the **shortest path** (fewest turns) to a goal state, which is desirable in a chatbot — students should reach an answer in as few exchanges as possible.
- Because the branching factor is small (5–8 intents) and the depth is low, BFS remains efficient here even though it can be memory-heavy on larger graphs.

In practice, Dialogflow does not run raw BFS — it uses a trained NLU (machine-learning) model to score every intent's similarity to the input and picks the highest-confidence match, which is closer to a **heuristic best-first search**, where the ML confidence score acts as the heuristic. The diagram below models intent selection as a BFS traversal, which is a useful simplified way to reason about and test the dialogue design before the ML model is trained.



Numbers show BFS visiting order: level by level, left to right

Figure 4: BFS-style level-order traversal used to model intent selection — the root query is expanded to all candidate intents (level 1) before any intent is expanded further into follow-up slot-filling states (level 2).

Justification of choice: BFS is chosen over Depth First Search (DFS) because DFS could commit early to one incorrect intent branch and dig deep into irrelevant follow-up questions before backtracking, giving a poor user experience. BFS instead compares all top-level intents fairly before going deeper, matching how Dialogflow itself first identifies the intent before entering intent-specific slot filling.

5. Testing and Evaluation

The agent was tested with a mix of clear queries, queries needing follow-up (slot filling), and unrelated/unknown queries to check the fallback behaviour.

User Query	Expected Intent	Result	Notes
"What time is my DSA class today?"	Timetable.Query	Pass	Correctly returned today's DSA class time.
"When is my exam?"	ExamSchedule.Query (+ slot filling)	Pass	Correctly asked "Which course?" before answering.
"Am I eligible for the AI exam?"	Attendance.Query	Pass	Returned attendance % and eligibility correctly.
"How do I add a backlog course?"	CourseRegistration.Query	Partial	Matched the general registration intent but did not distinguish between semesters.
"What's the weather today?"	Default Fallback Intent	Pass	Correctly identified as unrelated and offered help topics.
"library timing pls" (informal spelling)	CampusServices.Query	Pass	Handled informal/short phrasing correctly.

Evaluation Summary

- **Accuracy:** 5 of 6 test queries (about 83%) were matched to the fully correct intent on the first try; the remaining query was partially correct.
- **Response quality:** Responses were short, direct, and included the specific data the student asked for (date, time, percentage) rather than generic text.

- **Unknown query handling:** The Default Fallback Intent correctly caught the unrelated "weather" question instead of forcing it into an incorrect intent.

Suggested Improvements

- Add more training phrases covering informal spelling, short-hand, and regional phrasing to improve robustness.
- Add a dedicated entity/intent branch for special cases such as backlog or re-registration requests, instead of grouping them under the general registration intent.
- Add a hand-off option to a human staff member after two consecutive fallback responses, so students are never stuck without help.
- Track real usage logs over a semester to retrain the model on genuine student phrasing and further improve accuracy.